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ECG Arrhythmia Detection and Classification using Deep Learning on the MIT-BIH Arrhythmia Database

**Interim report on
Deep Learning Project
[ICT-4442]**

Submitted By

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INTERIM REPORT

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Title of the Project: ECG Arrhythmia Detection and Classification using Deep Learning on the MIT-BIH Arrhythmia Database.

GitHub Repository Link: <https://github.com/ishaankarmokar/ecg-arrhythmia>

1. Literature Review

Ten papers were chosen due to their relation to one of our four models or to the evaluation protocol. These papers have been summarised in Table I. The *Dataset/protocol* column states whether the test patients were also seen in training (intra-patient) or kept separate (inter-patient), since that is the most influential factor affecting results.

TABLE I. SUMMARY OF REVIEWED LITERATURE

Paper (Author, Year)	Method	Dataset / protocol	Key Result	Relevance to Project
de Chazal et al., 2004 [3]	Linear discriminants on ECG morphology + RR-interval features from two leads	MIT-BIH, 44 non-paced records; inter-patient (defines DS1/DS2)	SVEB Se 75.9%, +P 38.5%; VEB Se 77.7%, +P 81.9%	Source of our split, AAMI class grouping, baseline filtering and RR features
Zou et al., 2022 [4]	Random Forest on beat features + a CNN-derived "segment label" context feature	MIT-BIH; inter-patient	Acc 0.96; F1 N 0.98, V 0.93, S 0.74	Closest classical counterpart to M1; shows RF needs context beyond one beat
Lakshminarayan & Basil, 2016 [5]	Step-wise wrapper feature selection; LDA/QDA and an ensemble of 20 small ANNs	MIT-BIH, 44 records, ~115 Hz; inter-patient DS1/DS2	ANN ensemble: SVEB F 85.45, VEB F 91.14	Few well-chosen handcrafted features can compete; guides M1 features
Kachuee et al., 2018 [6]	1D residual CNN (5 residual blocks); features transferred to MI detection	MIT-BIH at 125 Hz, 5 AAMI classes; intra-patient (random beat split)	93.4% average accuracy	CNN design reference for M2; score not comparable with inter-patient testing
Wang et al., 2021 [7]	CWT scalogram + 2D CNN, with 4 RR features fused at the dense layer	MIT-BIH MLII, N/S/V/F; inter-patient DS1/DS2	PPV 70.75%, Se 67.47%, F1 68.76%	Supports joining RR context to CNN features, as in M2-M4
Rajpurkar et al., 2017 [8]	34-layer residual CNN labelling single-lead rhythm sequences	~64k Zio-patch records, ~29k patients; patient-disjoint test	Seq. F1 0.776 vs 0.719 for cardiologists	Deep CNNs generalise to new patients when trained on many patients
Salvi, 2024 [9]	9-layer CNN + BiGRU hybrid with LIME explanations	PhysioNet 12-lead rhythm database (45,152 patients), V1 lead, 5 rhythm classes balanced to 1,700 each; random 80:20 split	Acc 94.44%, F1 94.35%	CNN-BiGRU hybrid that motivates M3; rhythm-level task on another database, so not comparable
Sigurthorsdottir et al., 2020 [10]	1D CNN blocks + BiGRU with attention, test-time augmentation	PhysioNet/CinC 2020, 12-lead; hidden test sets	Score 0.511 on validation vs 0.167 on hidden test	BiGRU on conv features; shows the drop on unseen data sources
Mousavi & Afghah, 2019 [11]	CNN encoder + bidirectional RNN seq2seq decoder; SMOTE oversampling	MIT-BIH, N/S/V/F; intra- and inter-patient	Inter-patient S: +P 92.57%, Se 88.94%	Context across neighbouring beats and resampling for imbalance (M4, future work)
Hu et al., 2021 [12]	Transformer encoder over 1D-conv wave embeddings with RR-adaptive beat windows; pre-trained on MIT-BIH SVDB	MIT-BIH; trained on DS1 + first 5 min of each DS2 record, tested on the last 25 min (patient-adapted)	F1 S 0.86, V 0.96	Reference design for M4; uses test-patient data, so not a pure inter-patient result

Comparison: After the analysis of Table I, it becomes clear that the following facts take place. Firstly, intra-patient experiments on MIT-BIH with the random classification of beats have 93-99% accuracy [6],[20], whereas inter-patient studies on MIT-BIH have S-class sensitivity equal to 76-89%, which means that the second type of experiments is less accurate [3],[7],[11]. Secondly, it becomes clear that the accuracy is not a proper indicator, because N class occupies 89% of DS2. In addition, all inter-patient experiments of high precision use some additional information other than a single beat, for example, the RR intervals [3],[5],[7], the RR tailored beat window [12], segment-level label context [4] and the neighboring beats sequence [11]. Thirdly, the superiority of deep learning over hand-selected features in inter-patient experiments is less evident than in intra-patient ones [5],[7]. Our project will prove the above three hypotheses through the comparison of conventional methods with three deep learning ones within one inter-patient experiment protocol.

2. Dataset Acquisition & Preprocessing

Source: The MIT-BIH Arrhythmia Database [1], [2] (48 half-hour two-lead recordings from 47 patients, 360 Hz, beat labels by two cardiologists) was downloaded from PhysioNet with the `wfdb` package (`src/download_data.py`). The four paced records (102, 104, 107, 217) were excluded, as AAMI EC57 [13] and de Chazal et al. [3] recommend, leaving 44 records. The MLII lead is used throughout (record 114 stores it in the second channel, which the loader detects).

- **Cleaning:** Baseline wander artefacts correction by subtraction of the output signals from cascaded median filters (200 ms and 600 ms) and low pass Butterworth filter (4th-order, zero-phase, 35 Hz) [3].
- **Segmentation:** A single window of 360 samples (1 second) that contains the annotated R-peak; 100,664 beats in total excluding all other windows and windows without RR context information.
- **Encoding:** Beat annotations encoded into five classes (N, S, V, F, Q) according to AAMI standard [13]; integer numbers (0 - 4) for the purpose of using sparse categorical cross-entropy loss function.
- **Normalization:** Individual z-scoring of each beat for removing the amplitude differences between patients and electrodes placements; normalization of four RR-related features (previous RR, next RR, mean of previous 10 RR and their ratio) according to training data distribution.
- **Split:** Inter-patient split from de Chazal et al. [3]; DS1 for training, DS2 (22 different patients) for testing. Six DS1 records (108, 114, 116, 124, 207, 223) per patient are excluded for the purpose of validation, early stopping and CNN hyperparameters optimization. Each DS2 record is used only once per model.
- **Augmentation:** Augmentation technique is not used in this phase. Imbalanced dataset problem is solved through class weighted loss function (Deep learning model) and Bootstrap weight (Random forest).
- **Verification:** All the three split files are different from each other; the count of records per class is validated using the annotations file; and finally the score of each model is correct as compared to the testing score in the notebook.

TABLE II. BEATS PER CLASS IN EACH SPLIT (AFTER PREPROCESSING)

Split	N	S	V	F	Q	Total
Train (16 DS1 records)	34,853	716	2,889	389	8	38,855
Validation (6 DS1 records)	10,979	227	899	25	0	12,130
Test DS2 (22 records)	44,228	1,836	3,220	388	7	49,679

The imbalance is severe: N makes up 89-91% of every split, and only 15 Q beats remain after removing paced records (none in validation), so Q results are reported but not interpreted. F beats are concentrated in a few records. Records 201 and 202 come from the same patient yet fall on opposite sides of the standard split; we keep the standard division so results stay comparable with [3]-[5], [7].

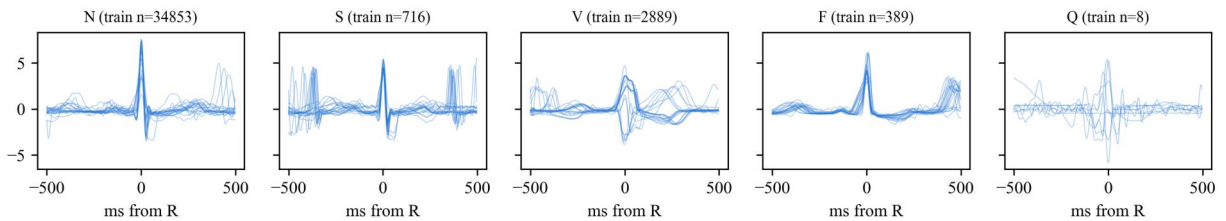


Fig. 1. Twenty random training beats per AAMI class after cleaning and per-beat z-scoring (x-axis: ms from the R peak). S beats differ from N mainly in timing, not shape.

3. Models Implemented So Far (all four of the proposed models)

All four models were tested on the same data split and measured with the same metric calculator (`src/evaluate.py`). Three runs of each deep learning model were performed using different seed numbers (42, 43, 44). The mean $\pm$ std was calculated due to the significant variance in terms of performance among separate sessions of the deep learning models. There are the same training procedures for the three deep learning models (`src/models/deep_common.py`): the RMSprop optimizer [16] (not Adam and not SGD), the inverse square root class balancing, the batch size 128,

the maximum of 30 training epochs, and the early stopping by the macro-F1 score on the validation N/S/V/F set (there is no Q beats in the validation set). All three deep learning models use the beat signal with four RR features.

- **M1 Random Forest [17] (Tanishq Sharma):** 55 features extracted per beat: db4 wavelet decomposition level-4 approximation coefficients, energy, std and peak for each detail band, 10 statistic properties (mean, spread, skewness, kurtosis, zero-crossing rate, ...) and the 4 RR features; 300 trees using class-weight based on sub-samples balanced by random undersampling N. Validation of this technique showed that undersampling N could be abandoned.
- **M2 1D CNN (Parth Upadhyay):** Conv1D-BatchNorm-ReLU-MaxPool with doubling number of filters, and followed by global average pooling and dense layers. According to the synopsis, the hyperparameters (learning rate, number of filters, kernel size, number of blocks, dropout and number of neurons in the dense layer) have been optimised by Differential Evolution [14] and Particle Swarm Optimisation [15] under the following conditions: 15 trainings for each case where the objective is validation macro-F1 score. Differential evolution yielded 0.452, particle swarm optimisation gave 0.447; DE configuration (lr=0.00124, filters=16, kernel=7, blocks=2, dropout=0.1, dense=32) is trained on entire DS1-train dataset. (In synopsis work plan, Optuna is mentioned; hyperparameters themselves are tuned by DE and PSO, as described in synopsis model description.)
- **M3 BiGRU [18] (Ishaan Karmokar):** The signal is downsampled to 120 samples (safe after the 35 Hz low pass) and fed to two bi-directional GRU layers (32 in each direction), and it is guaranteed that at any stage, the representation of the signal will contain the P wave before QRS and the T wave after it.
- **M4 Transformer encoder [19] (Riddhi Puneyani):** 90 patches of the signal are fed to the embedding layer using strided convolutions, d_model = 32, with positional encodings and then fed to two encoder layers with 4-head self-attention and GELU feedforward.

TABLE III. STATUS OF THE FOUR MODELS (DS2 TEST, MEAN $\pm$ STD OVER 3 SEEDS)

Model	Owner (Member)	Status	Preliminary Metric (macro-F1)	Notes
M1 Random Forest	Tanishq Sharma	Implemented, trained, evaluated	0.371 $\pm$ 0.001	V recall 0.78; S recall 0.01; F recall 0.00
M2 1D CNN (DE-tuned)	Parth Upadhyay	Implemented, trained, evaluated	0.360 $\pm$ 0.056	V recall 0.94; S recall 0.12; F recall 0.03
M3 BiGRU	Ishaan Karmokar	Implemented, trained, evaluated	0.333 $\pm$ 0.010	V recall 0.83; S recall 0.03; F recall 0.01
M4 Transformer encoder	Riddhi Puneyani	Implemented, trained, evaluated	0.392 $\pm$ 0.033	V recall 0.86; S recall 0.20; F recall 0.00

TABLE IV. COMPARISON ON DS2 UNDER THE COMMON PROTOCOL (MEAN $\pm$ STD OVER 3 SEEDS)

Model	Accuracy	Macro-F1 (5 classes)	95% CI (patient bootstrap)	Macro-F1 (N/S/V/F)	Rec. N	Rec. S	Rec. V	Rec. F
M1 Random Forest	0.939 $\pm$ 0.000	0.371 $\pm$ 0.001	[0.339, 0.418]	0.463 $\pm$ 0.001	1.00	0.01	0.78	0.00
M2 1D CNN (DE-tuned)	0.893 $\pm$ 0.041	0.360 $\pm$ 0.056	[0.313, 0.405]	0.449 $\pm$ 0.070	0.93	0.12	0.94	0.03
M3 BiGRU	0.891 $\pm$ 0.005	0.333 $\pm$ 0.010	[0.279, 0.376]	0.416 $\pm$ 0.012	0.94	0.03	0.83	0.01
M4 Transformer encoder	0.904 $\pm$ 0.031	0.392 $\pm$ 0.033	[0.333, 0.423]	0.489 $\pm$ 0.043	0.94	0.20	0.86	0.00
M2 1D CNN (untuned default)	0.927 $\pm$ 0.011	0.386 $\pm$ 0.048	[0.341, 0.404]	0.482 $\pm$ 0.060	0.97	0.16	0.90	0.00

The confidence interval resamples the 22 DS2 *patients* (2,000 bootstrap draws, `src/bootstrap_ci.py`), not individual beats, because beats from one patient are not independent.

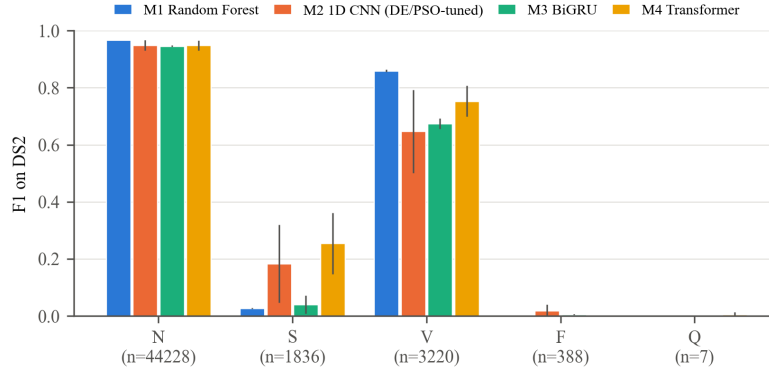


Fig. 2. Per-class F1 on DS2 (bars: mean of 3 seeds; whiskers: std). Support per class is shown under each label.

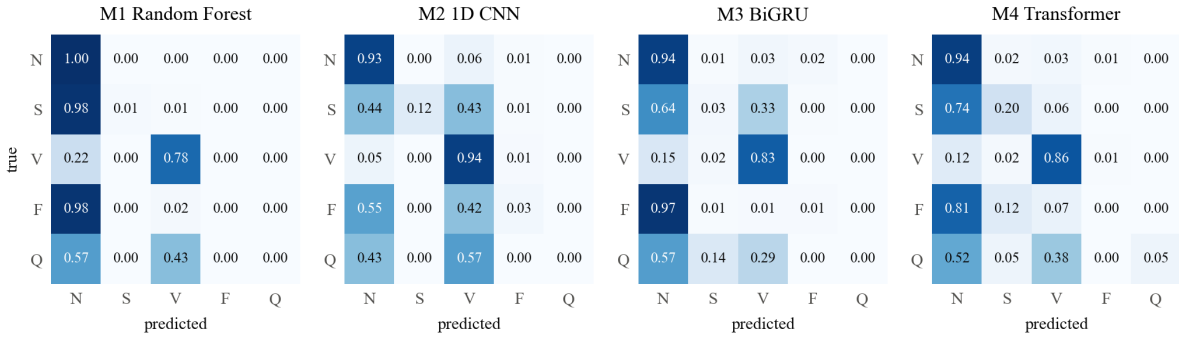


Fig. 3. Row-normalised DS2 confusion matrices, summed over the three seeds.

Preliminary observations. On the five-class macro-F1 measure, the Transformer encoder has the best current performance (0.392 ± 0.033), then come CNN without fine-tuning, the Random Forest, the fine-tuned CNN and the BiGRU. The differences between deep learning models are less significant than variability caused by random seeds, and 95% confidence intervals for all four models cross each other in patientbootstrapping (see Table IV). As the number of patients in the test set equals to 22, the order of these models is thus statistically indistinguishable. The Random Forest has the best accuracy (0.939) and the best V-class F1-score (0.859), as well as the highest stability for different seeds. This is due to correct prediction of N beats; the S recall for the Random Forest is only 0.014, F recall – 0. All deep learning models trade some of N and V precision for the improved S recall; the Transformer encoder recognizes the largest number of S beats (recall 0.197, F1 0.254). V class beats have relatively similar wide and abnormal QRS complexes and thus are easy to classify correctly for all models (recall 0.78-0.94). The main problem of all four models is detection of S and F beats on new patients. [3], [7].

Error analysis. Fig. 3 demonstrates an example of misclassification, where most of the S beats (64-98%) are assigned as class N by the RF, BiGRU, and Transformer networks. On the other hand, 43% errors are made by the CNN classifier by assigning S beats as class V. Tests classes show large skewness as well: record 232 has 1,381 out of 1,836 beats of DS2 class S (75%), record 213 has 362 out of 388 beats of class F (93%), and 372 out of 389 beats of training class F belong to record 208. Thus, the metrics S and F characterize the performance of the model on one or two patients, respectively, and F is learned from one patient.

Why the deep models behave this way. The optimal epoch was the first epoch or the second epoch for 9 out of 12 deep model architectures. After that, the training error continued to decrease, while the validation macro-F1 error continued to increase. Hence, the learning of the morphological features of the 16 training subjects became faster beyond this point. The validation performance does not indicate the test set performance as well. While the BiGRU achieved the highest validation macro-F1 performance of 0.462, it obtained the worst test performance.

The DE/PSO optimization enhanced the validation performance of the CNN from 0.423 to 0.433, but did not do the same for DS2 (0.360 ± 0.056 tuned vs 0.386 ± 0.048 default).

4. Individual Contribution Log (to date)

Member Name	Reg. No.	Task(s) Completed	Github ID
Ishaan Karmokar	230953162	Data download and shared preprocessing pipeline (config, filtering, segmentation, split); M3 BiGRU; notebooks 00 and 03; literature rows [9], [10]; report integration	ishaankarmokar
Parth Upadhyay	230953328	M2 1D CNN; DE and PSO hyperparameter search module and tuning runs; notebook 02; literature rows [6]-[8]	Parth-Upadhyay
Riddhi Puneyani	230953580	M4 Transformer encoder; shared deep-model training loop (RMSprop, class weights, early stopping, seeds); notebook 04; literature rows [11], [12]	riddhipuneyani
Tanishq Sharma	230953534	M1 Random Forest and feature extraction; common evaluation module and comparison tables/figures; notebooks 01 and 05; literature rows [3]-[5]	Tani_shar

5. Risk / Plan for Remaining Work

All four proposed models are already implemented. The remaining work is to improve their behaviour on unseen patients and to complete the final analysis and report. Table V lists the main risks and how we plan to handle them; Table VI gives the timeline.

TABLE V. RISKS AND MITIGATION FOR THE FINAL PHASE

Risk	Severity	Impact	Mitigation
Poor S and F recall on unseen patients	High	All models miss most S and F beats on DS2	Training-only augmentation (amplitude, noise, time shift); focal loss / SMOTE vs class weights; neighbouring-beat context for BiGRU and Transformer [11]
Weak model-selection signal	High	Six validation patients did not predict DS2; DE/PSO gains on validation did not carry over	Grouped leave-patients-out cross-validation on DS1 for early stopping and a larger DE/PSO budget; DS2 untouched until final runs
Overlapping confidence intervals	Medium	No model is yet statistically better than another	Keep patient-bootstrap CIs and 3-seed mean $\pm$ std in every comparison; avoid ranking claims the data cannot support
Classes too small or concentrated	Medium	Q has 7 test beats; S and F are dominated by records 232 and 213	Report per-record results and state these limits explicitly
Compute	Low	Full CPU run takes about 2.5 hours	Longer searches on Google Colab using the existing notebooks

TABLE VI. TIMELINE TO THE FINAL SUBMISSION

Work item	Owner	Target
Grouped (leave-patients-out) cross-validation on DS1; repeat DE/PSO with a larger budget on a Colab GPU	Parth Upadhyay	10 Oct
Data augmentation and imbalance study (focal loss, SMOTE on training data only) in the shared trainer	Riddhi Puneyani	12 Oct
Multi-beat context input (neighbouring beats) for the BiGRU and Transformer	Ishaan Karmokar, Riddhi Puneyani	15 Oct
Random Forest: patient-normalised features and decision thresholds calibrated on validation	Tanishq Sharma	15 Oct
Final training runs, per-record error analysis, comparison tables	All members	20 Oct
Final report (Part C, one methodology subsection per owner), repository clean-up	All members	24 Oct
Presentation (26-29 Oct); report + GitHub link (31 Oct)	All members	26-31 Oct

Declaration on the use of AI tools

An AI assistant (Claude, Anthropic) was used during this phase to help structure and debug the Python code, to locate and cross-check the bibliographic details of the reviewed papers, and to draft parts of the text of this report. All reported numbers are produced by the code in the project repository and can be regenerated with `run_all.sh`.

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